

3D Measurement System for Wound Care

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ABSTRACT

Wounds can be challenging to treat and often require periodic measurements to determine progression. Healthcare providers typically measure wounds (e.g., pressure injuries, venous ulcers, and diabetic foot ulcers) using rulers or images; however, these measurements tend to be subjective and inaccurate. We are developing a system that measures and tracks wound progression from 3-dimensional (3D) scans. In this research, we are evaluating our 3D software in comparison to wound measurement methods commonly used by providers. Our 3D software may facilitate the measurement, documentation, and visualization of wounds to support clinicians' treatment decisions and improve patient outcomes.

Index Terms: Visualization—3D modelling—3D scanning—Simulation—Measurement—Wound care

1 INTRODUCTION

Wound measurements are essential for determining wound progression and the efficacy of treatment. Healthcare providers commonly measure wounds using rulers and 2-dimensional (2D) images; however, these methods have limitations. While the ruler method is quick, simple, and inexpensive, its measurements are rudimentary and inaccurate [6, 7]. Furthermore, the ruler method requires direct contact with the wound, which may increase the risk of contamination, tissue damage, and discomfort [9]. The 2D image method has the advantage of being non-contact, but images cannot capture body curvature or wound depth and are prone to warping [3]. We are developing a system that computes wound measurements (i.e., length, width, depth, perimeter, and area) using 3-dimensional (3D) scans to facilitate monitoring wound progression and enhance visualization.

1.1 Proposed Research

We are investigating the ability of 3D software measurement methods to promote accuracy and early detection of healing progression. We are also exploring the development of custom simulated wounds via 3D modelling to replicate a wound progression.

1.2 Related Work

Sarp et al. proposed an AI-based model that utilizes the concept of a digital twin to track and predict wound healing [8]. Malone et al. demonstrated that 3D measurements may be important prognostic indicators of wound progression [4]. Jørgensen et al. identified that 3D area measurements may provide a more sensitive indicator of wound healing compared to 2D area [2].

2 OWN RESEARCH

2.1 3D Wound Measurement Software

We developed a software prototype that takes wound measurements from 3D scans. The software was developed in Unity 3D using the programming language C#. The software requires the input of a wound 3D scan. The software is independent of the 3D scanning

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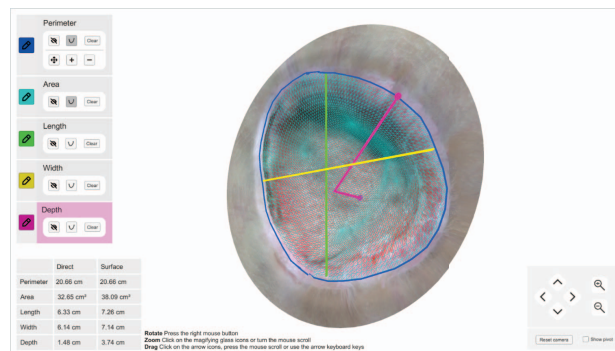


Figure 1: The user interface of our 3D wound measurement software.

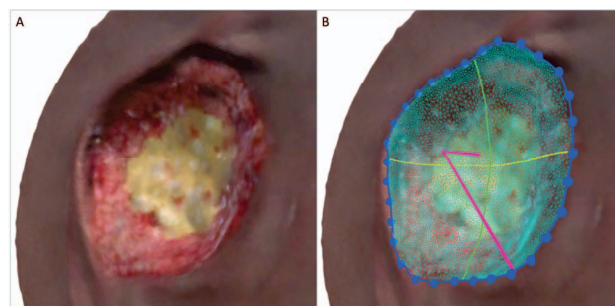


Figure 2: A 3D scanned simulated wound (A) without and (B) with measurements taken by our software.

technology; therefore, users can use a commercial 3D scanner, mobile device (e.g., phone or tablet), or photogrammetry software to generate the 3D scan. The software measures the length, width, depth, perimeter, and area of the wound from the 3D scan in a semi-automatic process. The wound measurements are visualized on the 3D scan. Figure 1 shows the user interface of the software. Figure 2 shows a 3D scanned simulated wound that has been measured in the software.

The software can implement guided and unguided methods to take wound measurements. In the guided method, measurements are automatically generated from user input for the perimeter measurement; and in the unguided method, the user manually indicates each measurement with minimal automation and restrictions. Moreover, the software can compute linear (i.e., length, width, and depth) and area measurements as direct or surface. Direct linear measurements are taken as the Euclidean distance between two points on the 3D model without accounting for the surface geometry, whereas surface linear measurements are taken as the geodesic distance on the non-parametric surface of the 3D model. Figures 3A and B show direct and surface linear measurements, respectively. Direct area is taken by triangulating the perimeter shape and computing the sum of the triangle areas, and surface area is measured as the sum of the mesh triangle areas enclosed in the perimeter. Figures 3C and D show direct and surface area measurements, respectively.

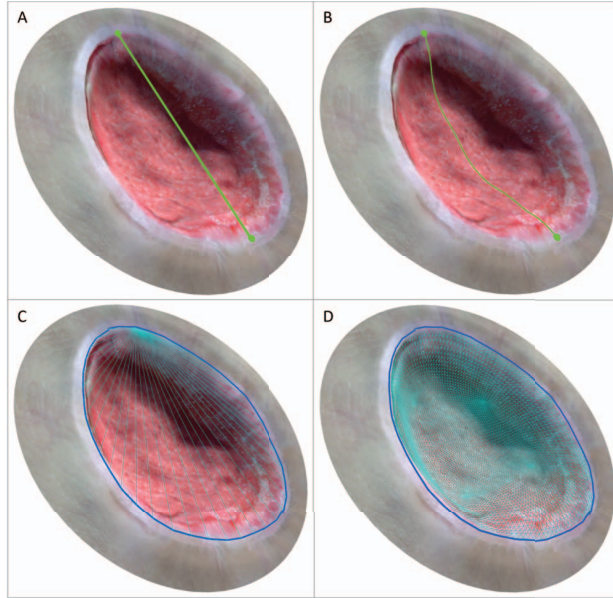


Figure 3: (A) Direct linear, (B) surface linear, (C) direct area, and (D) surface area measurements in our software.

2.2 Research Study

We conducted a study with participants at the New Jersey Institute of Technology to evaluate the inter-rater reliability and variability of physical (i.e., digital caliper), 2D image, and 3D software measurement methods [5]. In this study, the participants were asked to measure four simulated wounds using the three measurement methods. We found that our 3D software increased inter-rater reliability of complex measurements (i.e., perimeter and surface area) compared to the physical and image methods. We also found that there was significant variability in measurements between the three methods.

3 RESEARCH OBJECTIVE

Our research questions:

1. Does guided software increase the accuracy of wound measurements compared to unguided software?
2. Are measurements along the 3D surface a better indicator of change in wound size compared to direct measurements?

4 PROPOSED METHOD

4.1 Proposed Experiment 1: Development of Simulated Wounds

We will develop simulated wounds representative of a wound progression to use in the subsequent studies. Using Maya software, we will develop 3D models of a large and small wound. We will then use blend shape deformers to simulate the progression from the large wound to the small wound. The simulated wound progression will be validated by healthcare providers.

4.2 Proposed Experiment 2: Guided vs. Unguided Methods

Healthcare provider participants will be asked to use our 3D software to measure the simulated wound progression virtually using the guided and unguided methods. We will compare participants' guided and unguided measurements to the ground truth measurements of

each wound to determine if the guided method promotes measurement accuracy. We expect that the guided method will result in more accurate measurements, as it limits subjectivity and variability.

4.3 Proposed Experiment 3: Surface vs. Direct Measurements

Healthcare provider participants will be asked to use our 3D software to measure the simulated wound progression virtually using direct and surface measurements. We will compare participants' direct and surface measurements to determine if surface measurements detect change in wound size more rapidly compared to direct measurements. We expect that the surface measurements will promote more rapid detection of progression, as these measurements account for changes along the surface of the wound bed.

5 CONCLUSION

The IEEE VR Doctoral Consortium will allow me to discuss my research proposal with experts in the field and enhance my experiments. I hope to gain more perspective on 3D modelling, simulation, visualization, and measurement techniques that can be applied to clinical wound care.

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